Improving Hedge Timing for Corn Producers Under Weak Predictive Signals: A Machine Learning Approach

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1 Introduction

Commodity futures markets play a very important role in risk management for agricultural producers and market participants. In particular, corn futures are widely used by producers to hedge systematic risk by selling contracts in advance to lock in future sale prices. From a hedging perspective, producers benefit from executing these contracts at the highest possible price, thereby securing more favorable revenue outcomes. This creates a natural incentive to consider whether the timing of hedge execution can be improved.

While futures markets are often assumed to be efficient in the short run, there is some evidence that short-horizon return patterns may exhibit some predictability. This raises an important economic question: can short-term predictability in corn futures returns be systematically exploited to improve hedging decisions?

To address this question, this study examines whether incorporating information from related financial assets, weather conditions, macroeconomic regimes, and commonly used quantitative trading indicators can improve the timing of hedge execution in corn futures markets. If expected returns over the next few days are negative, a producer may benefit from hedging immediately to lock in a higher price, thereby reducing hedging costs. Conversely, if expected returns are positive, delaying hedge execution may be advantageous. Even modest improvements in timing can translate into meaningful economic gains, given the scale and frequency at which these markets operate.

This paper contributes to the literature in two ways. First, we implement a range of machine learning models to forecast short-term corn futures returns, with a particular focus on nonparametric methods such as k-nearest neighbors (KNN), which are well-suited to capturing weak and local predictive structure in noisy financial data. Second, we introduce a simple, out-of-sample hedge-timing framework that maps return forecasts into economically interpretable decisions, allowing model performance to be evaluated in terms of realized hedge execution prices rather than purely statistical metrics.

The results suggest that, while predictive signals are weak and traditional accuracy metrics show only marginal differences across models, the KNN-based approach generates economically meaningful improvements in hedge execution. In particular, applying the model to a simple timing rule yields a positive average

hedge gain and a steadily increasing cumulative improvement relative to naive benchmark strategies. These findings indicate that even modest predictive signals, when translated into decision rules, can have tangible value for producers operating in large and frequently traded commodity markets.

2 Literature Review

The predictability of agricultural commodity returns has been a subject of debate in financial economics for quite a long time. Arguments centered around the efficient market hypothesis suggest that all available information is reflected in current prices (Fama, 1970). However, recent literature in agricultural economics and machine learning indicates that frictions, information lags, and cross-market dependencies can create short-term predictive opportunities. In the past, corn price analysis relied heavily on linear econometric models such as ARIMA or GARCH. However, as computational power has increased, focus has shifted toward machine learning methods capable of capturing the nonlinear relationships often associated with demand shocks.

A primary reference in this field is the work by Gu, Kelly, and Xiu (2020), "Empirical Asset Pricing via Machine Learning," published in the Review of Financial Studies. While their work focuses on equities, their methodology is a gold standard for the exact models used in our project. They demonstrate that while simple linear regressions fail to capture risk premiums, techniques like Lasso and Elastic Net significantly improve OOS performance by reducing the overfitting issue. Our approach differs from Gu et al. by applying these methods to the highly seasonal and fundamentals-driven corn market. While they focus on firm-specific characteristics, we utilize exogenous factors like weather data and yield curve dynamics. Furthermore, where their study uses a standard K-fold CV, we implement a 10-split Time-Series CV to carefully maintain chronological ordering and avoid look-ahead bias.

Furthermore, weather is perhaps the most significant fundamental driver of corn prices. In the American Journal of Agricultural Economics, Roberts and Schlenker (2013) explored the non-linear effects of weather on crop yields and prices. Their research highlighted that extreme heat, rather than just average temperature, drives price volatility. We expound upon this economic intuition by including a "temp-prcp interaction" term in our feature set. This allows our models to capture the compounding effect of heat and drought, which is a significant improvement over models that treat temperature and precipitation as independent variables. Additionally, while Roberts and Schlenker focus on annual yield impacts, our study focuses on the immediate short-term price response (t+1), providing a more useful tool for active hedgers, as the objective states.

Finally, the "financialization" of commodities has led to a tighter link between corn, energy, and macro-indicators. A study by Tang and Xiong (2012) in Quantitative Economics notes that, after 2004, the correlation between agricultural goods and oil increased dramatically due to index investing, in which individuals aim to replicate the performance of a specific market benchmark. This justifies our inclusion of Crude Oil

and Soy futures as primary features in our dataset. Our methodology extends this by incorporating market regime signals via yield-curve metrics such as the 10-year minus 2-year spread and its daily changes. Many economic papers focus on these as recession indicators but fail to include short- and mid-term yield-curve-based indicators of recession. Those additional inclusions allow our model to identify short-term recessionary scares as well, not just long-term.

While existing pieces of literature use regularized regressions for asset pricing, the impact of extreme weather on yields, and the increasing correlation between corn and energy markets, there is limited research that combines all of these factors into a single predictive model. By focusing specifically on t+1 return predictability, we move beyond the broad annual forecasting common in agricultural economics to create insights for daily hedging and execution timing.

3 Model

3.1 Behavioral Assumptions

Market participants are heterogeneous and differ in their objectives. In particular, commercial hedgers such as agricultural producers are assumed to prioritize risk reduction rather than return maximization. Their decisions to hedge are therefore driven by operational needs, including production cycles, inventory management, and revenue stabilization, rather than short-term price forecasts.

As a result, hedging activity may exhibit systematic patterns over time, particularly around planting and harvest periods. These patterns can introduce predictable variation in futures demand that is not solely driven by new information about fundamentals.

In contrast, speculators are assumed to be more return-seeking but are limited in their ability or willingness to fully arbitrage away these patterns. This allows short-lived and weak predictability in returns to persist, particularly at short horizons.

3.2 Informational Assumptions

At each point in time t, market participants are assumed to have access to publicly available information, including financial variables, macroeconomic indicators, and observable weather data. The empirical model uses this same information set to form predictions about short-term futures returns.

While markets are generally efficient, this framework allows for the possibility that prices do not fully or instantaneously incorporate all available information. As a result, short-horizon returns may exhibit weak and short-lived predictability. Any such predictability is expected to be limited in magnitude and unstable over time, reflecting the inherently noisy nature of financial markets.

3.3 Institutional Assumptions

Corn futures are traded in organized, liquid markets that facilitate risk transfer between commercial hedgers and speculators. Contracts are standardized and centrally cleared, allowing participants to take positions with relatively low transaction costs and limited counterparty risk.

Market participants are able to adjust their hedge positions at a daily frequency, consistent with the short-horizon predictions generated by the empirical model. While trading frictions such as transaction costs and bid-ask spreads exist, they are assumed to be small but non-zero, and are incorporated into the evaluation of hedging performance.

3.4 Connecting Theory to the Empirical Model

Based on these ideas, we model the problem as predicting tomorrow's corn futures return using information available today. In particular, we write:

$$r_{t+1}^{\text{corn}} = f(\mathbf{X}_t) + \varepsilon_{t+1} \tag{1}$$

Here, r_{t+1}^{corn} is the one-day return on corn futures, and $\mathbf{X}_t$ represents the set of independent features observed at the end of day t. The term ε_{t+1} captures the unpredictable component, including random noise or unexpected shocks.

We do not assume a specific functional form for $f(\cdot)$, which links today's information to tomorrow's return. Instead, we adopt a flexible approach that allows the data to determine this relationship. This enables us to consider a wide range of models, including both linear and nonlinear methods. This flexibility is important because there is no strong theoretical guidance on the exact relationship between predictors and returns. Moreover, in financial data, especially at short horizons, it is often difficult to distinguish true nonlinear patterns from noise.

3.5 Hedge-Timing Decision Framework

To evaluate whether predictive signals have economic value, we translate return forecasts into a simple hedge-timing decision problem. Consider a corn producer who uses futures contracts to hedge price risk by taking a short position. For such a producer, a higher execution price when initiating the hedge is preferred, as it locks in more favorable revenue.

We construct a stylized one-period decision framework in which the producer must choose whether to hedge immediately at time t or delay the hedge by one day and execute at time $t+1$. Let P_t denote the futures price at time t, and P_{t+1} the realized price at time $t+1$.

We first define two naive benchmark strategies:

- **Impatience Rule (Benchmark 1):** Hedge immediately at time t, resulting in an execution price of

P_t.

- **Patience Rule (Benchmark 2):** Always delay and hedge at time $t+1$, resulting in an execution price of P_{t+1}.

We then define a model-based hedge-timing rule using predicted returns:

- If the predicted return $\hat{r}_{t+1} < 0$, hedge immediately at P_t
- If the predicted return $\hat{r}_{t+1} \geq 0$, delay and hedge at P_{t+1}

This rule reflects the intuition that if prices are expected to decline, the producer should hedge immediately to lock in a higher price, whereas if prices are expected to increase, delaying the hedge may be beneficial.

Let the model-implied execution price be defined as:

$$P_t^{\text{model}} = \begin{cases} P_t & \text{if } \hat{r}_{t+1} < 0 \\ P_{t+1} & \text{if } \hat{r}_{t+1} \geq 0 \end{cases} \tag{2}$$

We evaluate the model by comparing it to each benchmark. The improvement in hedge execution price relative to a benchmark is defined as:

$$\text{Gain}_t = P_t^{\text{model}} - P_t^{\text{benchmark}} \tag{3}$$

We summarize performance using the average gain per decision, the fraction of positive gains, and the cumulative gain over time:

$$\text{Cumulative Gain}_T = \sum_{t=1}^{T} \text{Gain}_t \tag{4}$$

Notably, this framework does not require high predictive accuracy. Instead, it evaluates whether even weak directional signals from $f(\mathbf{X}_t)$ are sufficient to improve hedge execution outcomes.

4 Data Description

The final dataset, `enriched_corn_data.csv`, is a daily time-series dataset constructed to study short-term predictability in corn futures returns. The data spans from October 1, 2000, to April 9, 2026, capturing approximately 26 years of daily observations. The unit of observation is a single trading day, and the dataset contains 6,384 observations across a feature set of 44 predictor variables and one outcome variable. The primary response variable, `corn_ret_t+1`, represents the next-day percentage return on corn futures prices (i.e., the return from day t to day $t+1$).

The dataset integrates multiple sources of data to capture the economic drivers of corn prices: commodity futures data (corn, crude oil, soybean) from Yahoo Finance; interest rate and macroeconomic data from

Federal Reserve Economic Data (FRED); weather data from the Meteostat API, which aggregates meteorological data from agricultural hubs across the Corn Belt; and production weights from the USDA NASS Quick Stats API, which are used to weight weather data by regional production importance. Together, these sources generate cross-asset variables, seasonality indicators, volatility features, weather controls, and macro-financial variables that comprise the dataset.

The mean of the dependent variable, `corn_ret_t+1`, is approximately zero (0.000290), consistent with standard asset pricing theory that short-horizon returns are difficult to predict and often dominated by noise. Its standard deviation, while relatively small (0.017810), is nontrivial, indicating meaningful day-to-day volatility in corn futures prices. Returns are approximately symmetric around zero, although the presence of large positive and negative shocks is reflected in the range of -23.6% to $+13.6\%$.

Variables `oil_close_roll_z` and `soy_close_roll_z` represent z-scores of crude oil and soybean futures prices over a rolling window, respectively. These are computed using closing price levels rather than returns. This choice is motivated by the potential for distortion introduced by forward-filling missing data. When returns are computed on forward-filled prices, sequences of artificial 0% returns can appear, biasing the mean and variance downward and weakening the integrity of the regression. Closing prices are preferred because they represent the market consensus price each trading day and are standard in financial analysis.

The heat stress proxy is formed as an interaction term between production-weighted temperature (`prod_w_temp`) and production-weighted precipitation (`prod_w_prcp`). Both underlying variables are computed as production-weighted averages of temperature and precipitation across major Corn Belt agricultural centers. Production-weighted temperature has a mean of 10.190576 and a standard deviation of 10.312085, while production-weighted precipitation has a mean of 1.780774 and a standard deviation of 3.590543. The interaction term is designed to capture joint variation in temperature and moisture conditions, rather than treating them as independent inputs. It has a mean of 24.787626 and a standard deviation of 63.004817, ranging from -95.43 to 776.21, reflecting the dispersion induced by combining the two underlying variables.

The Moving Average Convergence-Divergence (`macd_(12,26,9)`) is a technical indicator derived from corn futures closing prices. Widely used in technical analysis, it serves as a momentum gauge that helps identify price trends and trading signals. It has a mean of 0.291078 and a standard deviation of 9.250907, with a wide range from -43.31 to 44.72, indicating substantial variation over time in the difference between short- and long-term exponential moving averages.

Volatility measures exhibit moderate variation across the sample. The 5-day realized volatility (`corn_rv_5d`) has a mean of 0.034444 and a standard deviation of 0.019995, while the 20-day realized volatility (`corn_rv_20d`) has a mean of 0.073066 and a standard deviation of 0.031737. Both variables reach values above 0.23 at their upper extremes. This suggests that volatility is persistent over time, with periods of elevated variation.

The Relative Strength Index (`rsi_14`) has a mean of 50.739356 and a standard deviation of 16.702007. This is consistent with a distribution centered around neutral momentum conditions and suggests no systematic bias toward overbought or oversold regimes in the sample.

Given the variation in scale across variables, standardization is required prior to model estimation to ensure comparability across predictors with differing units and magnitudes, particularly for models sensitive to feature scale.

5 Econometric/ML Specification

To estimate the conditional return function $f(\mathbf{X}_t)$, we implement and compare a range of linear and nonlinear models, each capturing different aspects of the data-generating process.

Linear Models: We begin with multiple linear regression (MLR) as a baseline:

$$r_{t+1} = \beta_0 + \boldsymbol{\beta}'\mathbf{X}_t + \varepsilon_{t+1} \tag{5}$$

To address multicollinearity and potential overfitting, we also estimate regularized variants including Ridge, Lasso, and Elastic Net regressions. These models impose penalties on coefficient magnitudes to improve stability and out-of-sample generalization. Feature selection is further refined using a forward selection procedure combined with time-series cross-validation, allowing us to identify a parsimonious subset of predictors that retain predictive signal.

Tree-Based and Ensemble Methods: To capture potential nonlinear relationships, we implement decision trees, random forests, and gradient boosting models (XGBoost). These models allow for flexible interactions between predictors and are capable of approximating complex functional forms.

However, given the relatively weak signal in short-horizon returns, deeper and more complex models tend to overfit. To mitigate this, we constrain model complexity through shallow trees, regularization, and early stopping.

Nonparametric Methods: We also consider nonparametric approaches, including k-nearest neighbors (KNN) and support vector regression (SVR). These models impose fewer structural assumptions and instead rely on local similarity in feature space.

KNN, in particular, estimates predictions as:

$$\hat{f}(\mathbf{X}_t) = \frac{1}{K} \sum_{i \in \mathcal{N}_K(\mathbf{X}_t)} r_{i+1} \tag{6}$$

where $\mathcal{N}_K(\mathbf{X}_t)$ denotes the set of nearest historical observations.

The number of neighbors K is selected using time-series cross-validation, balancing bias and variance.

Model Selection and Evaluation: All models are estimated using a strict out-of-sample evaluation framework designed to respect the time-series structure of the data. The dataset is split into training and

test sets using an 80–20 chronological split, where the final 20% of observations are held out as a test set to evaluate true out-of-sample performance.

Within the training set, hyperparameters are selected using 10-fold time-series cross-validation. Specifically, we use a rolling (forward-chaining) cross-validation scheme, where each fold preserves temporal ordering and ensures that training data always precedes validation data. This approach avoids look-ahead bias and more accurately reflects the real-time forecasting problem.

Hyperparameters such as regularization strengths, tree depth, and the number of neighbors K (for KNN) are chosen to minimize cross-validated root mean squared error (RMSE).

6 Results

6.1 Predictive Performance Across Models

Table 1: Out-of-Sample Predictive Performance Across Models

Model	**RMSE**	**MAE**	R^2
KNN Regression	**0.017579**	**0.012115**	**0.005106**
Random Forest	0.017619	0.012111	0.000567
Lasso Regression	0.017628	0.012097	-0.000480
Elastic Net Regression	0.017628	0.012097	-0.000480
Ridge Regression	0.017630	0.012123	-0.000649
XGBoost Regression	0.017631	0.012130	-0.000799
Radial SVR	0.017636	0.012104	-0.001388
Regression Tree	0.017647	0.012177	-0.002641
Linear SVR	0.017665	0.012186	-0.004616
Forward-Selected MLR	0.017678	0.012238	-0.006103
MLR (Reduced Multicollinearity)	0.017705	0.012287	-0.009221
MLR (Full Multicollinearity)	0.017759	0.012328	-0.015372

Table 1 presents the out-of-sample performance of all models evaluated in this study. Across specifications, predictive accuracy is highly compressed, with RMSE values ranging from 0.01758 to 0.01776. These differences are economically negligible, indicating that no model provides a meaningful improvement in forecasting accuracy over others.

Consistent with this, R^2 values are close to zero or negative for nearly all models. This implies that most specifications perform no better than, and in many cases worse than, a naive benchmark that predicts the unconditional mean return. Even the best-performing models, such as KNN and Random Forest, achieve only marginally positive R^2, suggesting that the signal-to-noise ratio in short-horizon corn futures returns is extremely low.

Taken together, these results reinforce a central feature of financial return prediction: increasing model complexity does not necessarily translate into improved predictive performance. Nonlinear models such as XGBoost and support vector regression do not outperform simpler approaches, and in some cases exhibit slight overfitting.

6.2 Model Comparison and Interpretation

While overall predictive performance is tightly clustered across models, examining model-specific behavior provides insight into how different approaches handle weak and noisy signals.

Nonparametric and ensemble-based methods such as KNN and Random Forest achieve marginally better performance than other specifications. The tuning results for KNN (Figure 1) show that cross-validated RMSE decreases sharply as the number of neighbors increases from very small values, before flattening out for larger K.

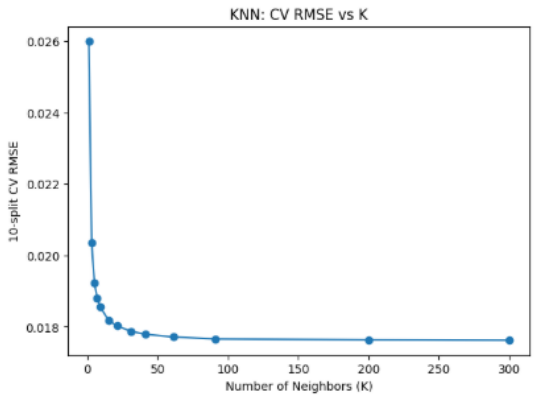

Figure 1: Choosing Number of Neighbors (K) using CV

This reflects a classic bias-variance tradeoff: small K values lead to high variance and overfitting, while larger K values stabilize predictions through local averaging. The plateau at higher K suggests that the model is primarily reducing noise rather than extracting strong underlying structure.A similar pattern is observed in Random Forest models (Figure 2), where increasing the number of trees beyond a moderate level produces negligible improvements in performance.

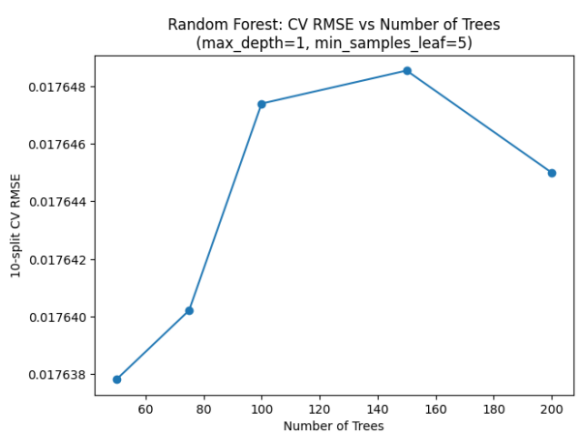

Figure 2: Choosing Number of Trees using CV

This indicates that ensemble averaging helps reduce variance, but does not uncover meaningful predictive

signal beyond a certain point.

Regularized linear models further reinforce the presence of weak signal. The Lasso, Ridge, and Elastic Net tuning plots (Figure 3) show that model performance improves as regularization is introduced, but quickly plateaus as coefficients shrink toward zero.

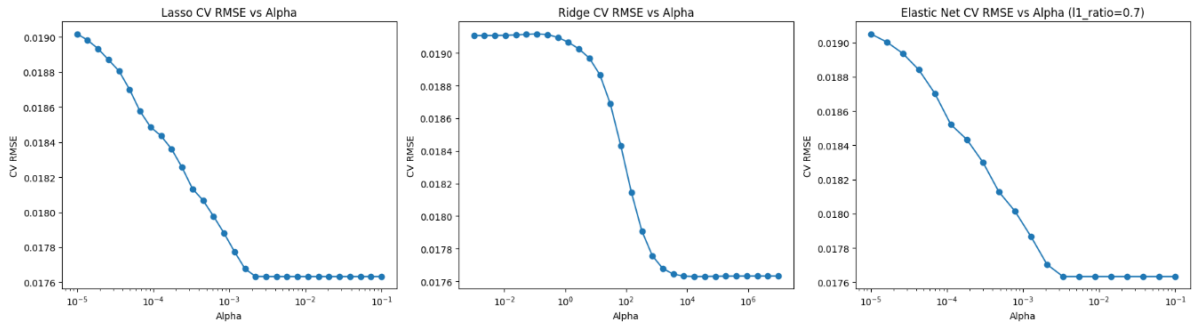

Figure 3: Model Improvement with change in λ

This suggests that only a small subset of features contributes marginal predictive value, while the majority of variables add noise.

Feature importance analysis from **XGBoost** (Figure 4) indicates that predictive signal is distributed across a handful of variables, including weather measures (e.g., temperature), yield-related variables, and market activity indicators.

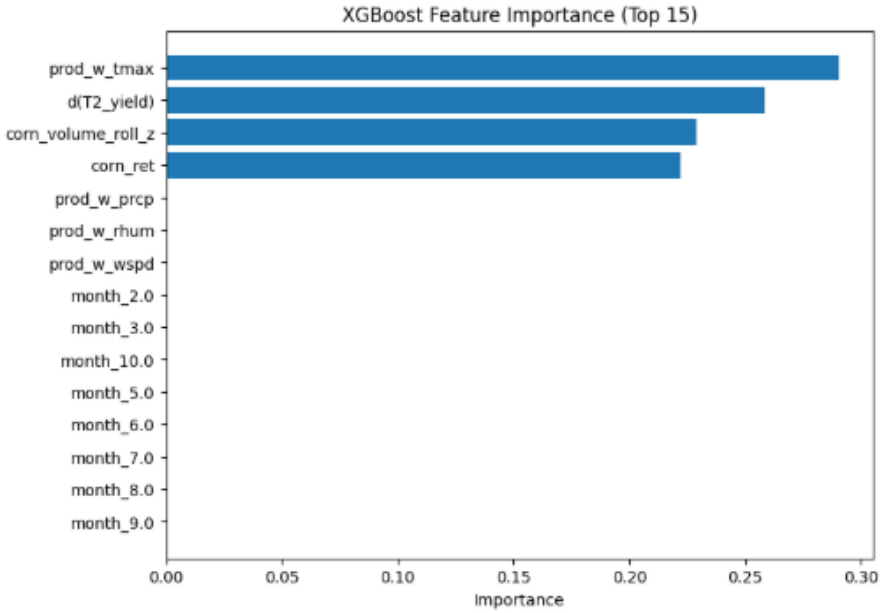

Figure 4: Individual Feature Importance for XGBoost

However, the absence of a dominant feature further supports the conclusion that predictive structure is diffuse and weak.

Despite their weaker statistical performance, linear models provide valuable economic interpretation. The **forward-selected MLR model** reveals several economically intuitive relationships (Figure 5). Returns exhibit mild autocorrelation, as higher lagged returns are associated with higher future returns, suggesting short-term momentum effects. The realized 5-day volatility is positively correlated with returns as well which is consistent with the idea that commodities experience sudden upward price spikes from bad news and gentle downward trends. This is opposite to how equities behave, where the result of bad news are downward price spikes and gentle upward recoveries.

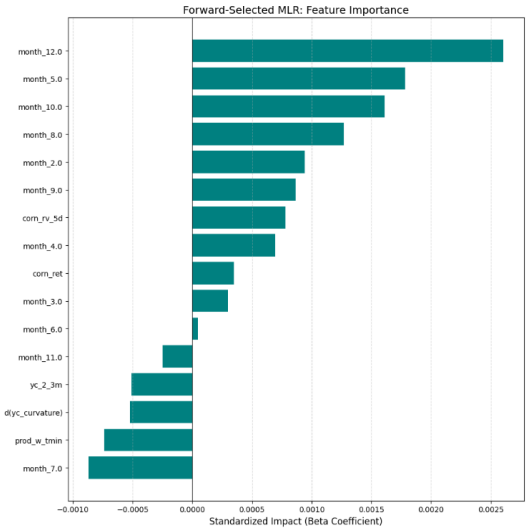

Figure 5: Individual Feature Importance for Forward-Selected MLR

Macroeconomic variables also play a role: a decrease in the short-term yield spread (2-year minus 3-month) is associated with higher corn futures returns, consistent with periods of increased recession risk where risk premia may rise. Similarly, increases in yield curve curvature are associated with lower returns, potentially reflecting improved macroeconomic conditions and reduced demand for risk compensation.

Weather and seasonal patterns are also economically consistent. Higher minimum temperatures and summer months, particularly July, are associated with lower returns, reflecting increased supply during harvest periods. Conversely, months such as December and May, which correspond to lower supply or key growing phases, are associated with higher returns.

Taken together, these results suggest that differences in model performance are driven primarily by variance reduction rather than the discovery of strong predictive relationships. Nonlinear and nonparametric methods provide slight improvements by smoothing noise, while linear models, although less accurate, offer interpretable insights into the economic drivers of returns.

6.3 Economic Value and Dynamic Performance of Hedge Timing

Although statistical differences across models are minimal, the economic value of predictions becomes more pronounced when evaluated within the hedge-timing framework. Using the KNN-based decision rule, we observe an average gain of approximately 0.167 per hedging decision relative to the immediate hedging benchmark, with a cumulative gain exceeding 200 over the evaluation period. When compared to a strategy that always delays hedging, the model achieves an even larger cumulative gain of approximately 327. Despite

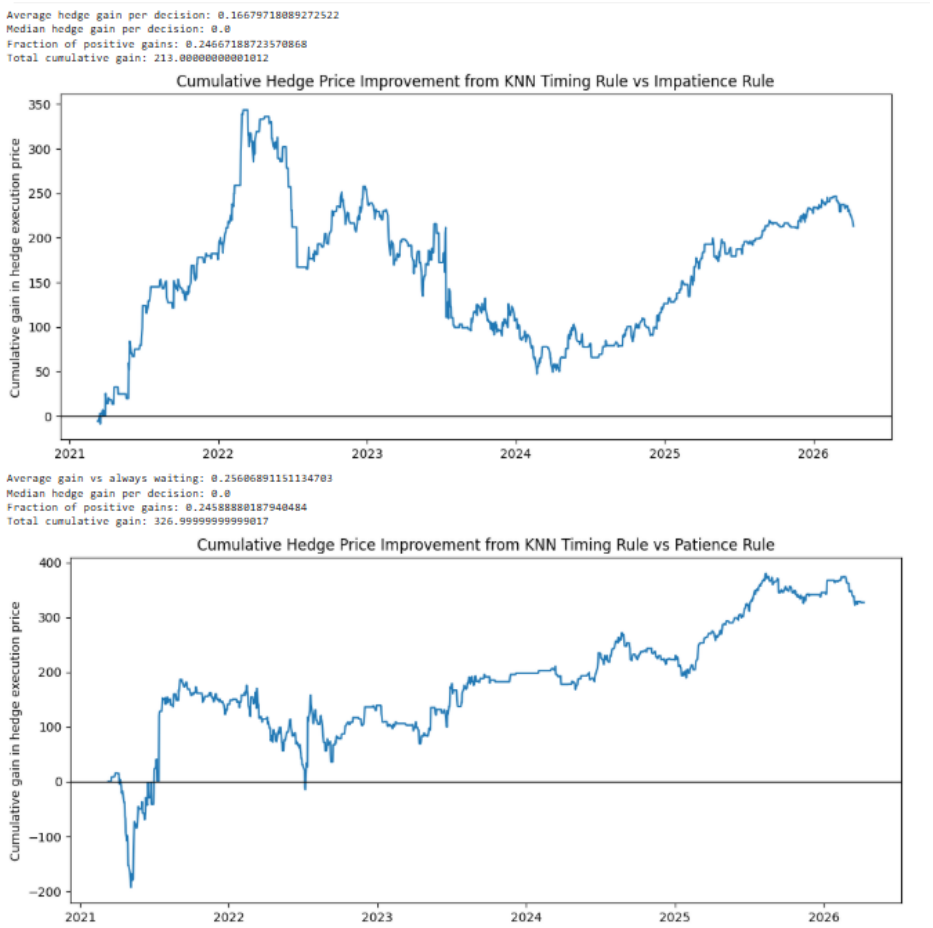

Figure 6: Cumulative Hedging Price Improvement for KNN Timing Rule vs Naïve Benchmarks

these positive aggregate outcomes, the distribution of gains is highly asymmetric. The median gain per decision is zero and the fraction of positive outcomes is relatively low (approximately 25%), indicating that most individual predictions do not generate gains. Instead, overall profitability is driven by a smaller number of correct directional forecasts that coincide with larger price movements.

Figure 6 illustrates the cumulative hedge price improvements over time. While performance exhibits periods of drawdown, particularly during volatile market conditions, the overall trajectory remains upward-sloping. This suggests that the model does not rely on a single market regime or isolated events, but instead accumulates gains gradually through time.

The presence of drawdowns further highlights the importance of evaluating performance over longer horizons. Short-term fluctuations may obscure the underlying economic value, but the persistence of positive cumulative gains indicates that even weak and infrequent predictive signals can translate into meaningful improvements in hedge execution.

Taken together, these results emphasize an important distinction: while predictive accuracy remains limited, the economic value of forecasts arises from their ability to improve decision timing in a small number of critical instances.

6.4 Key Takeaways

Several key insights emerge from the results:

- Predictive performance is tightly clustered across models, with negligible differences in RMSE and near-zero R^2 values.
- Increasing model complexity does not meaningfully improve forecasting accuracy, reflecting the weak signal in short-horizon returns.
- Nonparametric methods such as KNN provide slight statistical improvements through variance reduction rather than strong signal extraction.
- When translated into hedge-timing decisions, even weak predictive signals generate economically meaningful gains.
- The economic value of the model arises from a small number of correct directional predictions rather than consistent accuracy.

7 Conclusion

The empirical results indicate that predictive signals for next-day corn returns remain notably weak, consistent with the weak-form efficient market hypothesis, and non-linear methods appear superior relative to traditional econometric approaches. KNN Regression emerged as the most robust model in terms of predictive accuracy, achieving the lowest test RMSE (0.017579) and the highest positive R-squared (0.0051). Random Forest also outperformed the linear benchmarks, yielding a test RMSE of 0.017619 and an R-squared of 0.00057.

In contrast, the standard Multi-Linear Regression (MLR) and Forward-selected MLR models yielded negative R-squared values on the test set (-0.0152 and -0.0061, respectively). Despite this low explanatory power, feature selection processes identified that yield curve curvature changes, realized volatility (corn_rv_5d), and certain weather-interaction variables provided the most consistent informational value. The success of KNN

and Random Forest suggests that these models are better fit to capture the local dependencies and non-linear interactions that linear models overlook.

There are several limitations to consider in our research paper. Firstly, we only look at daily prices, meaning our model predicts what will happen tomorrow based on how the market closed today. In reality, corn prices move every second. In theory we could miss a large price swing that occurs in the middle of the day, i.e. intraday volatility from significant weather news.

Another limitation to consider is model stability over time, which is always liable to change. The relationships between predictors and corn futures returns may evolve due to changes in market structure or broader macroeconomic conditions. For instance, shifts in the behavior of speculative capital or changes in the role of index investment could alter the relevance of certain predictors. As a result, models trained on historical data may not generalize well to future periods, particularly if structural breaks occur. This raises concerns about the long-run reliability of the estimated relationships and suggests that continuous model updating would be necessary in practice.

Future research could improve on this framework by shifting the prediction target from next-day returns to slightly longer horizons, such as 5-day or 10-day returns. These mid-term horizons may contain a stronger and more stable signal, since very short-term returns are often dominated by noise. In addition, the problem could be reframed as a classification task rather than a regression problem. Instead of predicting the exact magnitude of returns, models would only predict the direction (positive or negative), which may be more reliable for guiding hedge timing decisions (and more feasible).

From a methodological perspective, the model could also be improved by better accounting for changing market conditions. One way to do this would be to incorporate regime-based approaches, such as spline models or similar techniques that allow relationships between variables to shift during periods of high volatility or macroeconomic stress. This would help capture structural breaks that standard models may miss. In addition, applying dimensionality reduction techniques like PCA to the macroeconomic and weather feature set could reduce noise in the data and help prevent overfitting, especially given the relatively large number of predictors.

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8 Appendix

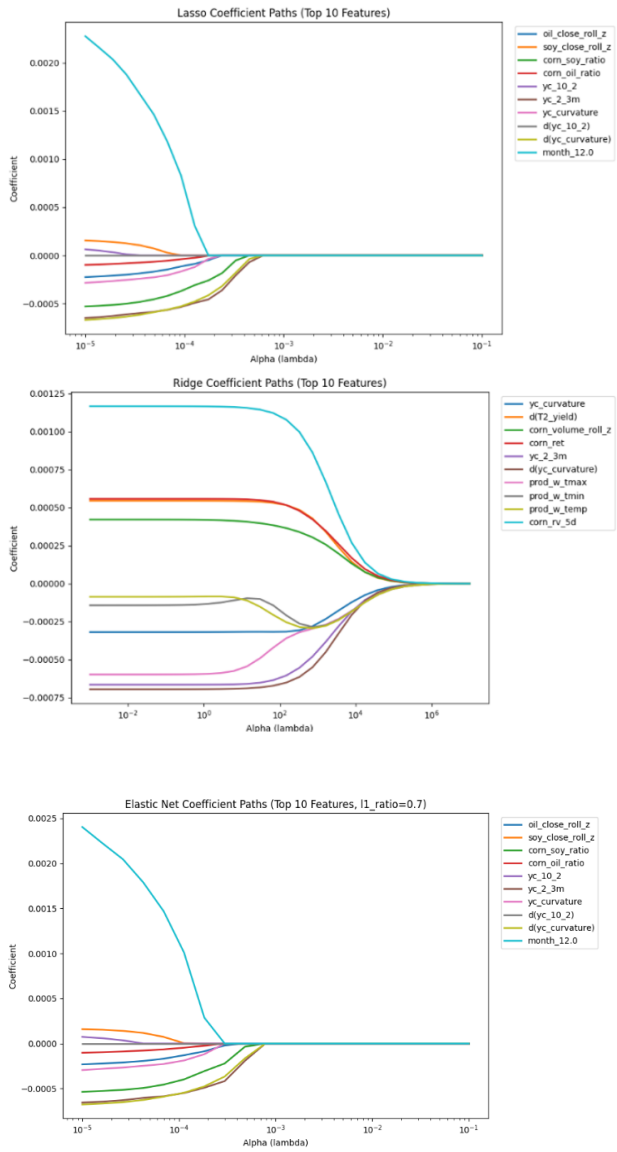

Figure 7: Coefficient Paths for Different Regularized Linear Models